

MANAN BANSAL

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EDUCATION

Dr. B. R. Ambedkar National Institute of Technology, Jalandhar

2024 – Present

B.Tech in Information Technology; CGPA: 9.40

Jalandhar, Punjab

Ch. Chhabil Dass Public School

2023 – 2024

CBSE Class XII; Percentage: 97.8%

Ghaziabad, Uttar Pradesh

TECHNICAL SKILLS

Languages: C++, Go, Python, JavaScript, SQL

Frontend: React, Next.js, Tailwind CSS

Backend & Systems: Node.js, Express.js, REST APIs

AI/ML: PyTorch, Transformers, LangChain

Databases: PostgreSQL, MongoDB, MySQL

Core CS: Data Structures & Algorithms, OOP Concepts, DBMS, Operating Systems, Computer Networks

PROJECTS

MockingJay-Claw – Agentic CLI Coding Assistant

TypeScript, Bun, OpenRouter

- Built a terminal-based **agentic AI coding assistant** with **3 execution modes** (Agent, Plan, Ask) and a **14-tool suite** (9 filesystem, 3 web, 2 skills), using Firecrawl for live web research.
- Designed a **staged in-memory filesystem overlay**: every file creation, edit, deletion, and shell command is queued, and **nothing reaches disk without explicit approval** via unified-diff review.
- Implemented a **tool-driven agent loop** that inspects the codebase, calls tools, and feeds results back into context; Plan mode emits schema-validated plans of **1–15 steps** with low/medium/high complexity tags.
- Enforced **4 safety guards**: path-traversal blocking, sensitive-path exclusion (`.env`, `.git`), file-read size caps, and approval-gated writes.

SIDER – Redis-Compatible In-Memory Data Store

Golang

- Built a **Redis-compatible in-memory data store** from scratch in Go, with a **single-threaded epoll event loop** over raw syscalls that multiplexes concurrent clients without per-connection threads.
- Hand-wrote a **RESP2 encoder/decoder** covering **all 6 wire types** and added **command pipelining** (multiple commands per round trip), so `redis-cli` and standard Redis clients work unmodified.
- Implemented **6 commands** (PING, SET with EX, GET, TTL, DEL, EXPIRE) with **hybrid key expiry**: passive deletion on access plus active sampling of **20 keys** per idle cycle, repeating while **>25%** are expired.
- Wrote protocol tests covering valid, malformed, and incomplete RESP2 requests.

Sahayak – AI Disaster Response Platform

React, Node.js, MongoDB

- Built a full-stack platform with **3 role-based dashboards** (citizen, admin, volunteer) automating a **5-stage pipeline**: emergency report, AI triage, admin review, volunteer assignment, and proof verification.
- Integrated **Gemini 2.5 Flash** to analyze incident reports and generate **severity scores with confidence values**, feeding a prioritized admin review queue.
- Implemented **proof-based mission closure**: volunteers upload completion photos to Cloudinary, and admins approve or reject them through a verification queue.
- Designed **12 REST APIs** and **6 Mongoose models** (reports, volunteers, tasks, verifications, notifications, users) with **JWT**-protected admin routes.

LEADERSHIP & ACHIEVEMENTS

The Cybernauts Society, NIT Jalandhar

2025 – Present

Student Coordinator

- Led end-to-end execution of **4 technical events** with **200+ participants**, overseeing problem design, event infrastructure, logistics, and on-ground operations.
- Coordinated the **technical and operations teams**, driving event workflows and resolving execution challenges to ensure smooth delivery.

Academic Achievements

- Ranked among the **top 5 students** in the B.Tech. batch of 2028 with a **9.40 CGPA**.
- Secured **Rank 1** in School with **97.8%** in CBSE Class XII.